

Piecewise-Defined & Implicit Functions



Evaluating piecewise functions

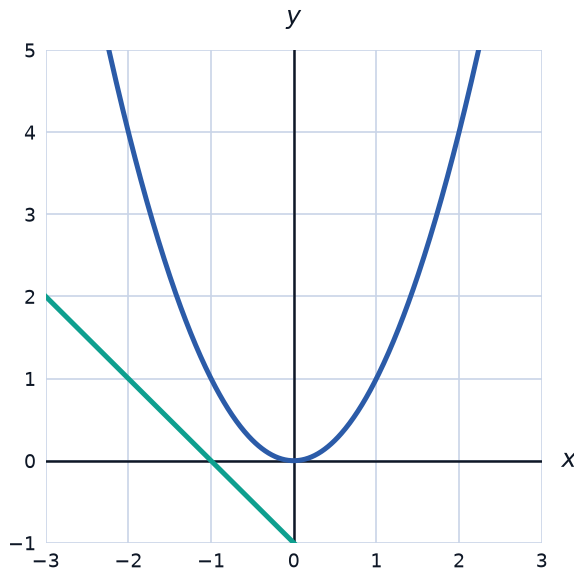
- 1 Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.
- 2 A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

Checking continuity of piecewise functions

- 3 Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.
- 4 Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

Visualizing a piecewise graph

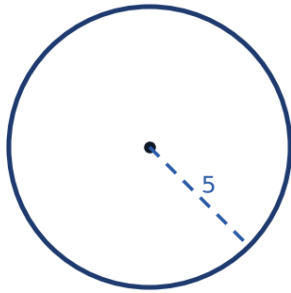
- 5 The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.



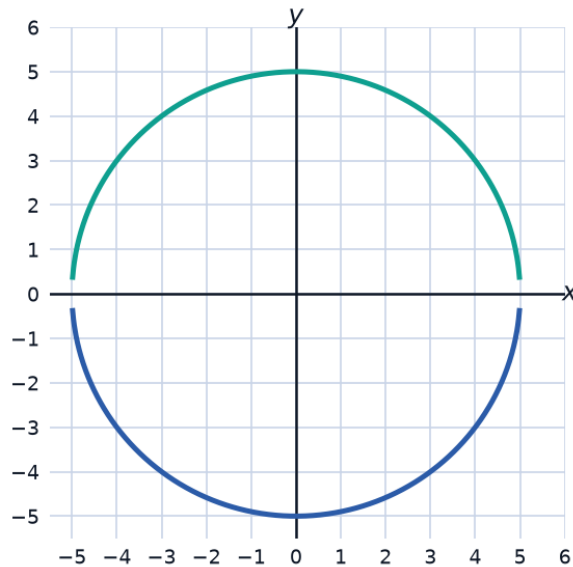
- a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there.
- b State the range of f shown on the graph.

Implicit relations and testing for functions

- 6 Determine whether $x^2 + y^2 = 25$ defines y as a function of x . Justify using the vertical line test.

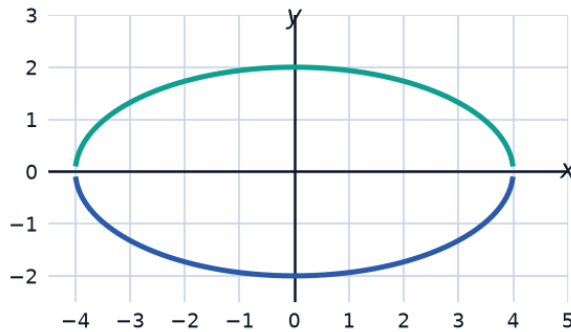


- 7 Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.



Implicit differentiation-style reasoning (algebraic slope)

- 8 For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y -values when $x = 2$, and describe what this means about the relation as a function.



Modeling with piecewise functions

- 9 A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x , and find the tax owed on an income of 25{,}000 dollars.

Piecewise functions with three or more pieces

- 10 A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.
- 11 Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

Graphing implicit relations point-by-point

- 12 For the implicit relation $y^2 = x + 4$, find all y -values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x -value.



Piecewise functions and average rate of change

- 13 For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.
- 14 Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.
- 15 A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

Multiple choice — easy 15

- 16 Given $f(x) = 3x - 5$ for $x < 0$, and $f(x) = -5x + 9$ for $x \geq 0$, find $f(-5)$.
- a. 34
 - b. -19
 - c. -20
 - d. -21
- 17 Given $f(x) = -2x - 5$ for $x < 3$, and $f(x) = 6x - 1$ for $x \geq 3$, find $f(2)$.
- a. 11
 - b. -9
 - c. -10
 - d. -8

- 18 Given $f(x) = -6x + 4$ for $x < 4$, and $f(x) = 6x + 5$ for $x \geq 4$, find $f(8)$.
- a. 53
 - b. 54
 - c. -44
 - d. 52
- 19 Given $f(x) = x + 2$ for $x < -5$, and $f(x) = -5x - 9$ for $x \geq -5$, find $f(-2)$.
- a. 1
 - b. -2
 - c. 0
 - d. 2
- 20 Given $f(x) = -3x + 3$ for $x < 1$, and $f(x) = -6x - 9$ for $x \geq 1$, find $f(1)$.
- a. -15
 - b. -14
 - c. 0
 - d. -16
- 21 Given $f(x) = -5x + 1$ for $x < -1$, and $f(x) = 3x + 6$ for $x \geq -1$, find $f(-1)$.
- a. 2
 - b. 3
 - c. 6
 - d. 4

22 Given $f(x) = 4x - 7$ for $x < 2$, and $f(x) = 6x - 5$ for $x \geq 2$, find $f(3)$.

a. 14

b. 5

c. 13

d. 12

23 Given $f(x) = 3x - 8$ for $x < -1$, and $f(x) = 2x$ for $x \geq -1$, find $f(-5)$.

a. -22

b. -24

c. -10

d. -23

24 Given $f(x) = 6x + 1$ for $x < -1$, and $f(x) = x - 2$ for $x \geq -1$, find $f(-5)$.

a. -29

b. -30

c. -28

d. -7

25 Given $f(x) = -5x + 5$ for $x < 5$, and $f(x) = 3x - 6$ for $x \geq 5$, find $f(6)$.

a. 12

b. 11

c. -25

d. 13

- 26 Given $f(x) = 2x - 7$ for $x < -1$, and $f(x) = -5x - 4$ for $x \geq -1$, find $f(-6)$.
- a. 26
 - b. -18
 - c. -19
 - d. -20
- 27 Given $f(x) = x - 9$ for $x < 5$, and $f(x) = 4x - 9$ for $x \geq 5$, find $f(0)$.
- a. -9
 - b. -6
 - c. -8
 - d. -10
- 28 Given $f(x) = -3x + 8$ for $x < -3$, and $f(x) = -2x + 6$ for $x \geq -3$, find $f(-7)$.
- a. 28
 - b. 30
 - c. 29
 - d. 20
- 29 Given $f(x) = 5x + 2$ for $x < -4$, and $f(x) = -2x + 3$ for $x \geq -4$, find $f(-7)$.
- a. -34
 - b. 17
 - c. -32
 - d. -33

30 Given $f(x) = -2x + 2$ for $x < 4$, and $f(x) = -x - 4$ for $x \geq 4$, find $f(-1)$.

- a. 4
- b. 5
- c. -3
- d. 3

Multiple choice — medium 15

31 Find the value of k that makes $f(x) = x + k$ for $x < 4$, and $f(x) = -x - 1$ for $x \geq 4$, continuous at $x = 4$.

- a. -10
- b. -8
- c. -9
- d. -1

32 Find the value of k that makes $f(x) = -2x + k$ for $x < 2$, and $f(x) = 7x - 1$ for $x \geq 2$, continuous at $x = 2$.

- a. 17
- b. 16
- c. 9
- d. 18

33 Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = 5x + 1$ for $x \geq 5$, continuous at $x = 5$.

- a. 67
- b. -14
- c. 66
- d. 65

- 34 Find the value of k that makes $f(x) = -9x + k$ for $x < -2$, and $f(x) = -7x + 6$ for $x \geq -2$, continuous at $x = -2$.
- a. 38
 - b. 2
 - c. 3
 - d. 1
- 35 Find the value of k that makes $f(x) = 7x + k$ for $x < 2$, and $f(x) = -5x - 8$ for $x \geq 2$, continuous at $x = 2$.
- a. -32
 - b. -31
 - c. -4
 - d. -33
- 36 Find the value of k that makes $f(x) = 3x + k$ for $x < 6$, and $f(x) = -5x - 8$ for $x \geq 6$, continuous at $x = 6$.
- a. -57
 - b. -20
 - c. -55
 - d. -56
- 37 Find the value of k that makes $f(x) = 3x + k$ for $x < -2$, and $f(x) = x - 9$ for $x \geq -2$, continuous at $x = -2$.
- a. -5
 - b. -6
 - c. -17
 - d. -4

- 38 Find the value of k that makes $f(x) = -9x + k$ for $x < -4$, and $f(x) = 2x - 6$ for $x \geq -4$, continuous at $x = -4$.
- a. -49
 - b. 22
 - c. -51
 - d. -50
- 39 Find the value of k that makes $f(x) = x + k$ for $x < 3$, and $f(x) = x - 5$ for $x \geq 3$, continuous at $x = 3$.
- a. -4
 - b. -5
 - c. -6
 - d. 1
- 40 Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = -2x + 2$ for $x \geq 5$, continuous at $x = 5$.
- a. 33
 - b. -48
 - c. 31
 - d. 32
- 41 Find the value of k that makes $f(x) = 7x + k$ for $x < 0$, and $f(x) = 8x - 9$ for $x \geq 0$, continuous at $x = 0$.
- a. -7
 - b. -9
 - c. -8
 - d. -10

- 42 Find the value of k that makes $f(x) = 9x + k$ for $x < 3$, and $f(x) = -x - 2$ for $x \geq 3$, continuous at $x = 3$.
- a. -33
 - b. -31
 - c. 22
 - d. -32
- 43 Find the value of k that makes $f(x) = -7x + k$ for $x < 3$, and $f(x) = 7x - 8$ for $x \geq 3$, continuous at $x = 3$.
- a. 35
 - b. -8
 - c. 33
 - d. 34
- 44 Find the value of k that makes $f(x) = x + k$ for $x < -1$, and $f(x) = 8x + 3$ for $x \geq -1$, continuous at $x = -1$.
- a. -4
 - b. -6
 - c. -5
 - d. -3
- 45 Find the value of k that makes $f(x) = 8x + k$ for $x < 5$, and $f(x) = -x + 5$ for $x \geq 5$, continuous at $x = 5$.
- a. 40
 - b. -41
 - c. -39
 - d. -40

Multiple choice — hard 15

46 For the implicit relation $x^2 + y^2 = 625$, find all y -values when $x = 24$.

a. $y = \pm 25$

b. $y = \pm 24$

c. $y = \pm 7$

d. $y = 7$

47 For the implicit relation $x^2 + y^2 = 289$, find all y -values when $x = 15$.

a. $y = 8$

b. $y = \pm 8$

c. $y = \pm 17$

d. $y = \pm 15$

48 For the implicit relation $x^2 + y^2 = 25$, find all y -values when $x = 3$.

a. $y = \pm 4$

b. $y = 4$

c. $y = \pm 3$

d. $y = \pm 5$

49 For the implicit relation $x^2 + y^2 = 841$, find all y -values when $x = 21$.

a. $y = 20$

b. $y = \pm 20$

c. $y = \pm 21$

d. $y = \pm 29$

- 50 For the implicit relation $x^2 + y^2 = 1681$, find all y -values when $x = 9$.
- a. $y = 40$
 - b. $y = \pm 40$
 - c. $y = \pm 41$
 - d. $y = \pm 9$
- 51 For $f(x) = x + 7$ on $x < 1$ and $f(x) = -4x + 2$ on $x \geq 1$, find the average rate of change of f on $[-3, 7]$.
- a. -4
 - b. -3
 - c. 3
 - d. -2
- 52 For $f(x) = 5x - 8$ on $x < 0$ and $f(x) = -3x + 7$ on $x \geq 0$, find the average rate of change of f on $[-1, 2]$.
- a. $\frac{14}{3}$
 - b. $-\frac{14}{3}$
 - c. $\frac{11}{3}$
 - d. $\frac{17}{3}$
- 53 For $f(x) = -5x - 8$ on $x < 2$ and $f(x) = 4x + 7$ on $x \geq 2$, find the average rate of change of f on $[-3, 7]$.
- a. $-\frac{14}{5}$
 - b. $\frac{9}{5}$
 - c. $\frac{14}{5}$
 - d. $\frac{19}{5}$

- 54 For $f(x) = -x + 8$ on $x < -3$ and $f(x) = 6x + 7$ on $x \geq -3$, find the average rate of change of f on $[-5, 0]$.
- a. $-\frac{6}{5}$
 - b. $-\frac{11}{5}$
 - c. $-\frac{1}{5}$
 - d. $\frac{6}{5}$
- 55 For $f(x) = 4x - 3$ on $x < -1$ and $f(x) = x - 6$ on $x \geq -1$, find the average rate of change of f on $[-2, 0]$.
- a. $\frac{7}{2}$
 - b. $\frac{5}{2}$
 - c. $-\frac{5}{2}$
 - d. $\frac{3}{2}$
- 56 A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.
- a. 32
 - b. 33
 - c. 31
 - d. $\frac{79}{2}$
- 57 A shipping company charges a flat 8 dollars for packages up to 3 kg, plus 3 dollars per kg for anything over 3 kg. Find the cost for a 11 kg package.
- a. 33
 - b. 41
 - c. 32
 - d. 31

- 58 A shipping company charges a flat 13 dollars for packages up to 2 kg, plus 2 dollars per kg for anything over 2 kg. Find the cost for a 7 kg package.
- a. 27
 - b. 24
 - c. 23
 - d. 22
- 59 A shipping company charges a flat 14 dollars for packages up to 1 kg, plus $\frac{5}{2}$ dollars per kg for anything over 1 kg. Find the cost for a 2 kg package.
- a. $\frac{33}{2}$
 - b. 19
 - c. $\frac{35}{2}$
 - d. $\frac{31}{2}$
- 60 A shipping company charges a flat 8 dollars for packages up to 2 kg, plus 3 dollars per kg for anything over 2 kg. Find the cost for a 9 kg package.
- a. 29
 - b. 30
 - c. 28
 - d. 35